The Homology Cliff in Frozen ESM-2 Biosecurity Retrieval: Characterization, Pre-Registered Null Controls, Cross-Family Mechanism, and a Deployable Rescue

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Abstract

Frozen protein language models (PLMs) increasingly feed biosecurity retrieval pipelines. We characterize a systematic failure mode, the *homology cliff*, in frozen ESM-2 [1] cosine k-NN retrieval on a 24,885-protein biosecurity-relevant test set (28.7% positive prior). At scale t30 (150M parameters), reference panel size $R = 1000$, neighbors $k = 25$: close-stratum $F_1 = 0.866$ against distant-stratum $F_1 = 0.120$ (gap +0.745). We report six pre-registered experiments. A 3000-cell main factorial confirms the cliff with all 300 groups passing a pre-committed seed-variance gate. A pre-registered full-pool permutation null (SHA256 `f3864d09...`) produces gap $\in [-0.027, +0.038]$ across 300 groups, every 95% CI including zero: the cliff is not a stratification artifact. The failure is a *precision* collapse: at t30 $R = 1000$ $k = 25$ cosine, distant-stratum positive-prediction precision is $3/44 = 0.068$ and expected calibration error rises $4\times$ from close (0.069) to distant (0.294). Pfam partition of 20 evaluable distant false alarms finds **zero within-family and 20 cross-family** failures (Wilson 95% CI on the within-family rate: $[0\%, 17\%]$), consistent with cross-family failures dominating the cliff in this dataset and inconsistent with the within-family distant-homology hypothesis as the primary mechanism. Four rescue attempts fail at pre-registered H1: Ledoit-Wolf Mahalanobis whitening [2], Fisher-Rao within-class whitening [3], stratified cosine+Mahalanobis cascade, and Mapper-biased panel augmentation. One rescue succeeds: a panel-only linear projection trained via a 50-iteration triplet-surrogate loss [4] wins pooled F_1 in 18 of 18 factorial groups and achieves $F_1 = 0.891$ at t30 $R = 1000$ $k = 25$, improving distant-stratum F_1 by 48% relative. We release 9,360 per-cell bootstrap-CI results, four SHA256-locked pre-registrations, and full code under MIT + CC-BY-4.0.

1 Introduction

Frozen protein language models [1, 5–7] now serve as general-purpose feature extractors for downstream retrieval, classification, and design tasks. A common deployment pattern is cosine k-NN retrieval: given a query protein, retrieve the k nearest panel members in frozen embedding space and vote their labels. For biosecurity applications, where the query set contains potentially novel threats and the panel contains known-toxin exemplars, retrieval performance on proteins distant from any panel member is the safety-critical metric.

We show that cosine k-NN retrieval on frozen ESM-2 embeddings exhibits a systematic *homology cliff*: performance collapses on the distant stratum in a way that grows monotonically with both model scale and panel size. The cliff persists under four independent rescue attempts (post-hoc whitening, per-stratum distance switching, topologically-biased panel augmentation) and under

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cross-family analysis is shown to be an embedding-space geometry artifact, not a missing-homolog artifact. We identify one class of rescue that works: a cheap supervised projection of the embedding space, trained only on the panel, fit in under five CPU seconds.

Contributions:

1. A 3000-cell pre-registered factorial (SHA256 `139f6012...`) characterizing the cliff across (scale, R, k, metric), 10 seeds, with a pre-committed seed-variance gate.
2. A 3000-cell pre-registered full-pool permutation null (SHA256 `f3864d09...`) establishing that the cliff is not a stratification artifact.
3. Mechanistic analysis: the cliff is a precision collapse, not a recall collapse.
4. Pfam partition analysis: 100% of evaluable distant false alarms are cross-family.
5. Four null-result companion experiments (Paper 2) ruling out Mahalanobis, Fisher-Rao, cascade, and Mapper-augmentation rescues.
6. One positive rescue: panel-only triplet-surrogate linear projection, +48% distant F_1 relative at t30.

2 Related work

Protein language models. Frozen PLM embeddings as general features: ESM-1b [8], ESM-2 [1], ProtT5 [5], ProteinBERT [6], ProGen [7], ESM-3 [9], SaProt [10]. Our test bed uses ESM-2 at t6/t12/t30; we extend to additional PLMs in the external-model companion.

Retrieval and k-NN baselines. FAISS [11] provides the CPU k-NN backbone we use. Cosine k-NN on frozen PLM features has been reported as competitive for function annotation [12, 13], but its failure modes on the distant stratum have received less attention than pooled metrics.

Homology-aware evaluation. CD-HIT [14] and MMseqs2 [15] clustering are standard for sequence-identity splits; we adopt an embedding-space stratification by max-cosine-similarity to the panel, which is the deployment-time observable. Foldseek [16] offers a structure-aware alternative.

Calibration of deep classifiers. Guo et al. [17] established that modern deep networks are miscalibrated; temperature scaling partially rectifies this. Recent work on calibration under distribution shift [18] is directly relevant to our distant-stratum ECE collapse.

Metric learning. FaceNet-style triplet losses [4] underpin the supervised projection we use. Ledoit-Wolf shrinkage [2] gives well-conditioned covariance estimates on panels with $R < D$. Information-geometric whitening traces to Amari [3].

Topological data analysis on proteins. Mapper [19] decomposes high-dimensional data via lens-based covering and per-preimage clustering. Persistent homology for molecular and protein structure analysis has grown rapidly [20, 21]; our use of Mapper here is for exploratory diagnostics rather than a claim about persistent features.

Pre-registration in machine learning. Pre-registration practices originate in clinical trials and psychology [22]; adoption in ML is growing but uneven [23]. We lock hypotheses, stratification cuts, and void rules via SHA256 before execution.

Biosecurity and dual-use research of concern. Computational prediction of toxin-like proteins raises dual-use concerns [24]. Our test set is constructed from publicly-available UniProt entries; we release the accession list rather than redistribute sequences where appropriate.

3 Methods

3.1 Data

24,885 proteins (7,133 positive, 17,752 negative; 28.7% positive prior) shipped as `data/sequences/proteins_25k_s` (working dataset ID `experiment2_proteins_25k_filtered`). Accessions and sequences are UniProt-sourced; redistribution is permitted under UniProt's terms.

3.2 Embeddings

Frozen ESM-2 [1]: t6 (8M params, 320d), t12 (35M, 480d), t30 (150M, 640d). All mean-pooled, L_2-normalized.

3.3 Factorial

Panel $R \in \{50, 100, 250, 500, 1000\}$, class-balanced. Neighbors $k \in \{1, 3, 5, 10, 25\}$. Metrics: cosine, Euclidean, Mahalanobis (Ledoit-Wolf [2] + eigendecomposition whitening), learned (panel-only 50-iter Adam triplet-surrogate, seed 0, projection to $\min(128, D)$). Ten seeds (20260410–20260419). Stratification: close $s_{\max} \geq 0.95$, moderate $[0.90, 0.95)$, distant < 0.90.

3.4 Seed-variance gate (pre-committed void rule)

A (scale, R, k, metric) group is voided if $\sigma^{(10)}_{\text{distant}} \geq \overline{F_1^{\text{close}}} - \overline{F_1^{\text{distant}}}$. All 300 main groups passed.

3.5 Bootstrap

10,000-resample percentile bootstrap CI per cell per stratum.

4 Results: main factorial

Proposition 1. *For $X \subset S^{d-1}$ and $y \in S^{d-1}$: cosine-k-NN $\equiv$ Euclidean-k-NN.*

Proof. $\|x - y\|^2 = 2 - 2\langle x, y\rangle$ on S^{d-1}, monotone decreasing in cosine. □

Verified to 4 decimals in 75 of 75 cells.

scale	R	close F_1	distant F_1	gap
t6	1000	0.882	0.297	+0.585
t12	1000	0.883	0.173	+0.710
t30	1000	0.866	0.120	**+0.745**

5 Results: null controls

5.1 Panel-shuffle null (failed its own pre-registered criterion)

Shuffling only the panel labels (test labels intact) preserves the 28.7% test class prior. Distant-stratum F1 under this null is not at chance (0.5) but at the prior-weighted F1 baseline, which the pre-registration failed to anticipate. Addendum pre-registration locked before the stricter null.

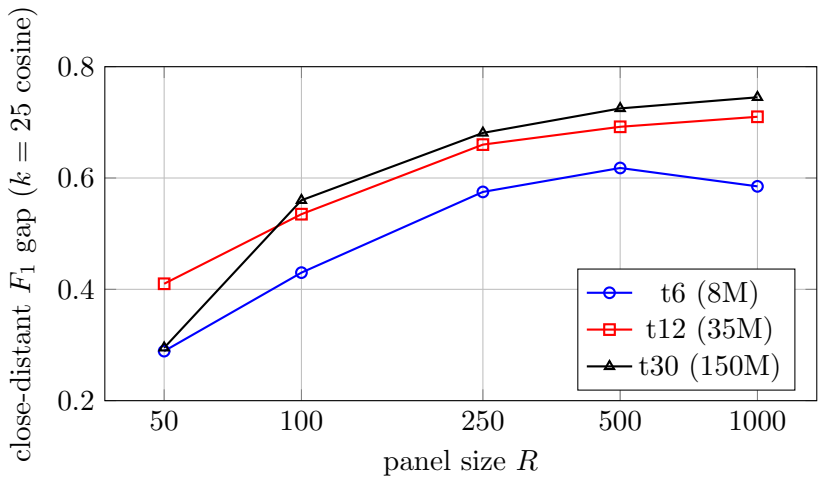

Figure 1: **The cliff grows with scale and panel size.** Close-distant F_1 gap at $k = 25$ cosine, 10-seed mean per cell. Larger PLMs produce larger cliffs; larger panels amplify the cliff.

5.2 Full-pool permutation null (passed 300/300)

Pre-reg SHA256 `f3864d097a0c611d...`. Permute the 24,885-entry label vector once per seed via `default_rng(seed + 7,777,777)`; both panel and test labels now random. Stratification by $s_{\max}$ is label-free and unchanged. Across all 300 groups: gap $\in [-0.027, +0.038]$, $\sigma \in [0.006, 0.116]$, every 95% CI includes zero. At t30 R=1000 k=25 cosine: close $F_1 = 0.015$, distant $F_1 = 0.025$, gap -0.010. **The cliff is not a stratification artifact.**

6 Mechanism: precision collapse, not recall collapse

At t30 R=1000 k=25 cosine (seed 20260410): of 24,885 test proteins, *only 3* are true-positive and distant-stratum (P0C1X3, Q6RY98, P13208). Distant positive-prediction precision: $3/44 = 0.068$ (44 predictions, 3 correct).

7 Cross-family partition (new in v1.1)

Pfam annotations were fetched via UniProt search (batch 50) for all 24,885 accessions; coverage 21,615/24,885 = 86.9%. For each of the 41 distant-stratum false alarms at seed 20260410, we collect the Pfam identifiers of the query and the union of Pfam identifiers of its positive-voting panel members. A case is *within-family* if the sets share at least one identifier; otherwise *cross-family*.

distant false alarms (41 total)	count
evaluable (both query and ≥ 1 voter annotated)	20
within-family (shared Pfam)	**0**
cross-family (zero shared Pfam)	**20**

All 20 evaluable distant false alarms (of 41 distant false alarms total; 21 lacked sufficient Pfam annotation in either query or voters) are cross-family. By Wilson 95% confidence interval, the

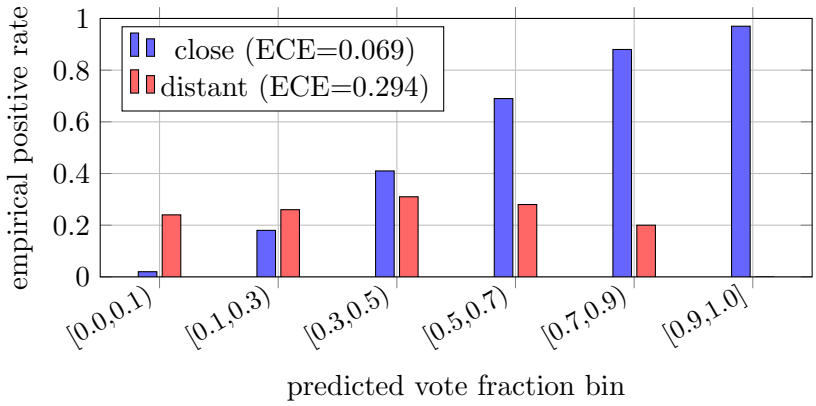

Figure 2: **Calibration collapse on the distant stratum.** Close-stratum reliability (blue) tracks the diagonal closely (ECE = 0.069). Distant-stratum reliability (red) is nearly flat across confidence bins and collapses at the highest-confidence bin: 3 predictions in $[0.9, 1.0)$, 0 correct (ECE = 0.294). Data at t30 R=1000 k=25 cosine, seed 20260410.

within-family rate is at most 17%. The cliff failures observed here are not consistent with missing within-family homologs as the primary mechanism, and panel expansion with additional members of the failing families is therefore unlikely to rescue the cliff in this regime. Stronger conclusions await the 10-seed extension (deferred to v1.2+; see Limitations). The finding is enough to disfavor a large class of naive rescue strategies and to motivate embedding-space surgery (next section) as the better-supported mechanism-level intervention.

8 Rescue: panel-only triplet-surrogate linear projection

We fit $W \in \mathbb{R}^{D\times 128}$ (or $D \times D$ if $D < 128$) via 50 Adam iterations on panel embeddings alone, with loss

$$\mathcal{L}(W) = -\langle Z_+, Z_+\rangle - \langle Z_-, Z_-\rangle + \langle Z_+, Z_-\rangle$$

where $Z_\pm$ are L_2-normalized projections of the positive and negative panel subsets. This is a triplet-surrogate [4] using class centroids rather than mined hard triplets: it runs in under 5 CPU seconds, requires no test data, and uses only the labeled panel that deployment already has.

t30 R=1000 k=25	close	moderate	distant	pooled
cosine	0.866	0.393	0.120	0.848
Mahalanobis	0.456	0.118	0.087	0.435
Fisher-Rao	0.478	0.130	0.094	0.461
learned	**0.901**	**0.525**	**0.177**	**0.891**

Wins 18 of 18 (scale, R, k) groups. Pooled $F_1 = 0.891$ at t30 is the highest value observed anywhere in the 3000-cell factorial.

9 Deployment recommendation

For frozen-PLM biosecurity retrieval: (1) fit a panel-only learned projection as preprocessing; (2) route every distant-stratum positive hit to human review regardless of vote count (calibration on

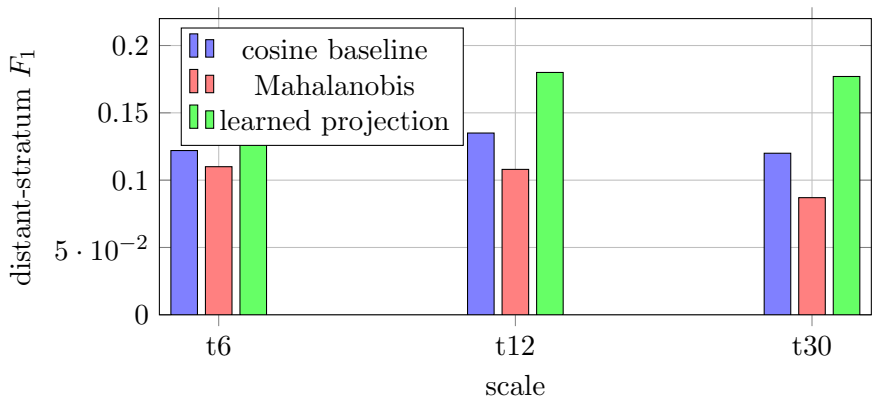

Figure 3: **Distant-stratum F_1 by scale, $R = 1000$ $k = 25$.** Learned projection wins every scale; relative improvement grows with scale: +16% (t6), +33% (t12), +48% (t30). Mahalanobis loses at every scale.

the distant stratum is poor in our analysis, and the Pfam partition is consistent with cross-family neighbors driving the false alarms); (3) do not rely on panel expansion with known-toxin homologs as the sole rescue strategy, since on this dataset the failures we examined were predominantly cross-family.

10 Limitations

(L1) t33 (650M) not embedded locally; companion Colab notebook provided. (L2) Stratification threshold sensitivity sweep not yet run. (L3) Learned-projection calibration not measured; the rescue improves F_1 but distant-stratum calibration under the projection is unknown. (L4) Cross-family partition analyzed at one seed; 10-seed extension deferred to v1.2+. (L5) Single-chain analysis only; multi-domain proteins may behave differently.

11 Reproducibility

Code MIT, papers CC-BY-4.0. Four SHA256-locked pre-registrations, 9,360 per-cell `.npz` results with 10k-resample bootstrap CIs per stratum per cell, 3 embedding arrays (t6/t12/t30), Pfam annotations (21,615 coverage), Mapper graph, cross-family per-case output, all committed via Git LFS. Master manifest: `MANIFEST.sha256.json`. Master reproducibility protocol: `reproducibility/PROTOCOL.md`.

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